

Software Requirements Specification (SRS)

GodsEye - AI-Based Missing Person Identification System

Version

1.0

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1. Introduction

1.1 Purpose

The purpose of this document is to specify the software requirements for GodsEye - an AI-Based Missing Person Identification System. The system enables law enforcement to securely manage missing person cases and perform AI-driven facial recognition on photos and CCTV video footage to identify missing individuals.

1.2 Scope

The GodsEye system provides 24x7 automated facial scanning and case management services, ensuring secure authentication and real-time detection through integration with InsightFace deep learning models and a MongoDB case database.

1.3 Intended Audience

- Police Officers
- System Administrators
- Software Developers
- Test Engineers
- Project Managers

2. Overall Description

2.1 Product Perspective

The GodsEye system communicates with a centralized MongoDB database and a Python AI microservice to process transactions and facial recognitions in real time.

2.2 Product Functions

- User Authentication (Login/Logout)
- Missing Person Registration
- Single Photo AI Detection
- CCTV Video Scanning
- Found Report Submission
- Case Status Management
- Dashboard Analytics
- System Diagnostics and Backup

2.3 User Classes

Police Officer

- Registers missing persons
- Scans photos and videos
- Submits found reports

Administrator

- Approves officer accounts
- Approves found reports
- Monitors system status and settings

3. Functional Requirements

FR-1 Login

- Accept user email and password.
- Verify credentials.
- Allow limited login attempts.
- Lock account after 5 incorrect attempts.

FR-2 Missing Person Registration

- Accept demographic details.

- Accept one or more facial photos.
- Verify exactly one clear face exists in each photo.
- Extract facial embeddings via AI.
- Save record and auto-generate case number.

FR-3 Single Photo Detection

- Accept field photo upload.
- Verify face presence.
- Compute cosine similarity against database.
- Return matches exceeding threshold.
- Display annotated bounding boxes.

FR-4 CCTV Video Scanning

- Accept video file upload.
- Sample frames at specified intervals.
- Perform detection on each frame.
- Record matches with exact timestamps.

FR-5 Found Report Submission

- Accept sighting details and location.
- Link report to a specific missing person case.
- Flag report as pending for admin review.

FR-6 Admin Review

- List pending found reports.
- Allow admin to approve or reject.
- Update case status to Found upon approval.

4. Non-Functional Requirements

Performance

- Login response < 2 seconds

- Single photo detection < 2 seconds
- Support multiple concurrent video scans

Security

- JWT based authentication
- Password hashing (Bcrypt)
- Role-based access control
- Transaction audit logs

Reliability

- 99.9% uptime for backend API
- Graceful failure if AI service is offline

Availability

- 24x7 service accessibility

Usability

- Simple sidebar navigation
- Clean visual feedback for detections

5. External Interface Requirements

User Interface

- Web browser and Desktop application
- Interactive data tables
- Video player with timestamp tracking

Hardware Interface

- Standard PC/Laptop
- Mouse and keyboard input

Software Interface

- MongoDB Database
- FastAPI AI Service
- Express Node.js Backend

Communication Interface

- REST APIs (JSON over HTTP)

6. Business Rules

- First registered user is granted Administrator rights.
- Officers must be approved before logging in.
- Match similarity must exceed the defined threshold.
- Cases cannot be closed without an approved Found Report.

7. Constraints

- System must operate with authorized accounts only.
- Internet connectivity is required to download AI models initially.

8. Assumptions

- Officers provide clear, front-facing reference photos.
- Uploaded videos are in standard formats (.mp4, .avi).

9. Use Cases

Use Case	Description
Login	Authenticate user credentials.
Register Case	Register a new missing person.
Scan Photo	Detect faces in a single field photo.
Scan CCTV	Scan a video file for missing persons.
Submit Report	Submit a found person sighting.
Approve Report	Admin verifies and closes a case.

1. Use Case: Login

Primary Actor: User

Preconditions: User has a registered active account.

Trigger: User submits credentials.

Main Success Scenario:

- User enters email and password.
- System verifies credentials.
- System issues session token.
- User views dashboard.

2. Use Case: Scan Photo

Primary Actor: Police Officer

Preconditions: User is logged in.

Trigger: User uploads an image.

Main Success Scenario:

- User selects photo.
- System extracts face vectors.
- System compares against database.
- Matches are displayed with bounding boxes.

10. Acceptance Criteria

- User authentication succeeds with valid credentials.
- Case registration correctly saves images and generates vectors.
- Face scanning correctly identifies registered persons above threshold.
- Found report approval updates case status to Found.
- All transactions are logged for auditing.

Problem Statement – GodsEye System

Law enforcement agencies struggle to process vast amounts of CCTV footage and field photos manually when searching for missing persons. Traditional methods are slow and prone to human error. An automated AI system addresses these challenges by enabling officers to accurately identify missing individuals using facial recognition 24x7. The objective is to provide a secure, reliable tool that processes images and video streams efficiently, managing the entire lifecycle of a missing person case from registration to closure, ultimately saving time and improving recovery rates.

Problem Definition – GodsEye System

Design and develop a secure, AI-driven Missing Person Identification System that enables law enforcement to perform high-speed facial recognition on field photos and CCTV videos to locate missing persons efficiently and accurately.